

Karthik Alluri

📍 Boulder, CO 📞 608-774-2444 ✉ kaal6482@colorado.edu 🔗 LinkedIn 🐙 GitHub 🌐 Portfolio

SUMMARY

Passionate AI/ML Engineer and Data Engineer with strong expertise in Deep Learning, Computer Vision, Data Processing, and NLP. Skilled in Python, TensorFlow, PyTorch, and scalable ML systems.

EDUCATION

University of Colorado, Boulder <i>Masters of Science in Computer Science(GPA: 3.910)</i>	May 2025 – Present Boulder, CO
Chaitanya Bharathi Institute of Technology <i>B.E. in Computer Science and Engineering</i>	Dec 2021 – Apr 2025 Hyderabad, India

EXPERIENCE

Cyient <i>Machine Learning Intern</i> - Engineered an intelligent semantic search assistant using the Gemini API and FAISS to handle complex natural language queries on extensive legal document corpuses. - Developed and optimized a custom embedding pipeline, fine-tuning vectorization processes to significantly enhance document retrieval precision and relevance for users. - Implemented advanced FAISS indexing strategies, achieving consistent sub-200ms query response times and ensuring high-performance scalability.	Jan 2025 – Apr 2025 Hyderabad
Wipro <i>Technical Intern</i> - Architected and built an AI-powered patient diagnostic platform that successfully reduced daily physician documentation time from hours to minutes, a 90% improvement. - Implemented robust FastAPI-TensorFlow ML workflows to enable real-time inference on sensitive clinical data, improving both diagnostic accuracy and processing speed. - Deployed automated data management and visualization modules that streamlined the analysis of patient records and supported clinical decision-making.	Jun 2024 – Aug 2024 Hyderabad
Enertech Comnet Pvt Ltd <i>Software Development Intern</i> - Spearheaded the front-end development of the client software's Version 2.0, implementing performance optimizations that improved overall UI responsiveness by 25%. - Engineered and integrated new UI components with backend REST APIs, enhancing data flow efficiency and enabling the deployment of scalable feature extensions.	Feb 2024 – Apr 2024 Hyderabad

PROJECTS

Aerodynamic Shape Optimization with GAN-Driven 3D Model Generation <i>AI Research Project</i> - Developed a GAN-based 3D model pipeline improving aerodynamic accuracy by 15% and reducing design time by 40%. - Integrated real-time CFD simulations achieving 12% drag reduction across optimized car models.	Jul 2024 – Feb 2025 Python, GANs, CFD
Automated Legacy Refactoring via Hybrid RAG & Multi-Agent Systems <i>Research Project</i> - Architected Multi-Agent RAG systems with self-correction loops, achieving a 92% code compilation success rate. - Engineered FAISS-based retrievers reducing hallucinations by 35% and cyclomatic complexity by 25% via automation.	Jan 2026 – Present Python, LangGraph, FAISS, LLMs (Claude)

TECHNICAL SKILLS

Languages: Python, SQL, C, JavaScript, HTML/CSS, Next.js

Machine Learning & GenAI: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, OpenAI API, Gemini API, RAG, FAISS, Pinecone, Chroma

Backend & Cloud: FastAPI, REST APIs, AWS (S3, IAM, SageMaker), Docker, GitHub Actions, AWS Bedrock

Frontend & Data: React, Next.js, Bootstrap, Power BI

Certifications: AWS Certified AI Practitioner, Google Cybersecurity Certificate